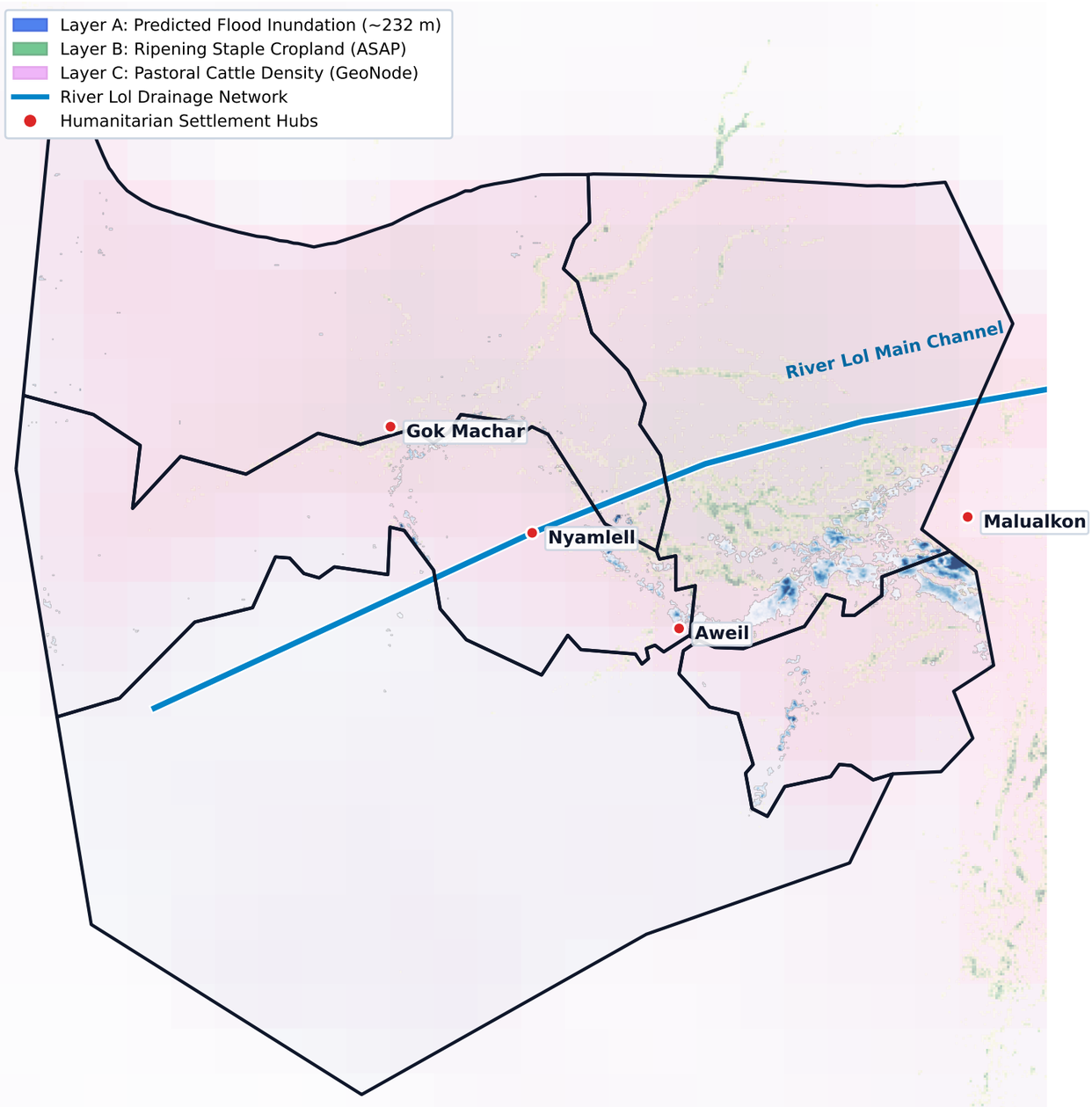
Visual 5: Multi-Layer Spatial Exposure & SLE Anticipatory Action Matrix

A. Multi-Layer Spatial Exposure Map (Peak Week 2024-11-04)



B. HUMANITARIAN EXPOSURE & SLE AUDIT

PHYSICAL ASSET EXPOSURE (PEAK SURGE 2024)

Stationary Cropland at Risk: 142 km² (34.2% of Aweil Arable Land)

Pastoral Cattle Herds at Risk: ~185,000 Heads (41.8% of Regional Herds)

SOCIETAL, LEGAL & ETHICAL (SLE) CONSIDERATIONS

[FOOD SECURITY] Stationary Crop Assets (Sorghum & Groundnuts):

- Vulnerability: Stationary crops cannot be relocated; submersion >48h triggers rot.
- Action Mandate: 2–3 week lead enables community-mobilized accelerated harvesting, bund construction, and grain silo elevation before access roads become submerged.

[LIVESTOCK PROTECTION] Mobile Pastoral Herds (Dinka Communities):

- Vulnerability: Herds can migrate, but false alarms provoke severe intercommunal conflict over grazing rights and expose cattle to endemic diseases in unfamiliar pastures.
- Action Mandate: 7–14 day high-confidence trigger enables orderly transhumance, preventative liver-fluke vaccination, and local dry-corridor coordination.